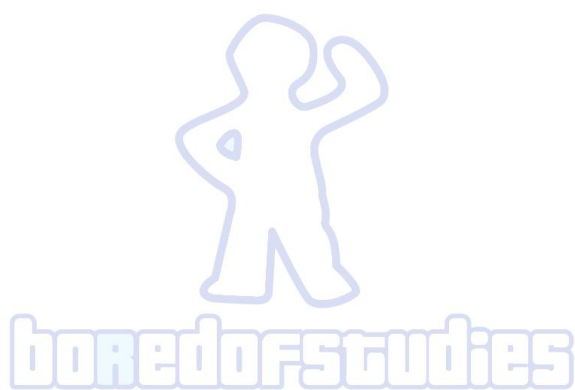




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2022

BORED OF STUDIES TRIAL EXAMINATION

4th October

Mathematics Extension 2

**General
instructions**

- Reading time – 10 minutes
- Working time – 3 hours
- Write using a black or blue pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

**Total marks:
100****Section I – 10 marks** (pages 2–4)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 5–10)

- Attempt Questions 11–16
- Allow about 2 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

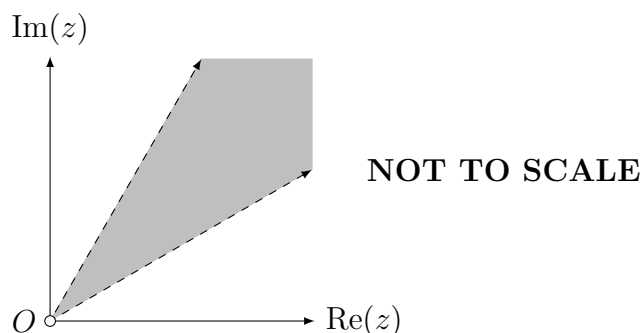
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 Which of the following is a unit vector that is perpendicular to $\frac{1}{4}\hat{i} + \frac{1}{2}\hat{j} + \frac{1}{2}\hat{k}$?

- (A) $\frac{2}{9}\hat{i} + \frac{1}{9}\hat{j} - \frac{2}{9}\hat{k}$ (B) $\frac{2}{9}\hat{i} - \frac{1}{9}\hat{j} - \frac{2}{9}\hat{k}$
(C) $\frac{2}{3}\hat{i} + \frac{1}{3}\hat{j} - \frac{2}{3}\hat{k}$ (D) $\frac{2}{3}\hat{i} - \frac{1}{3}\hat{j} - \frac{2}{3}\hat{k}$

- 2 Which of the following best represents the shaded region on the complex plane below?



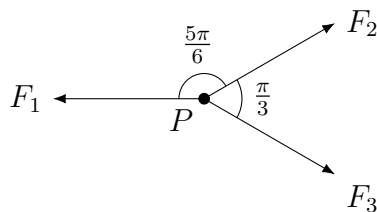
- (A) $\frac{\pi}{6} < \arg(z) < \frac{\pi}{3}$ (B) $\frac{\pi}{3} < \arg(z) < \frac{2\pi}{3}$
(C) $\frac{1}{2} < \frac{\operatorname{Im}(z)}{\operatorname{Re}(z)} < 3$ (D) $\frac{1}{2} < \frac{\operatorname{Re}(z)}{\operatorname{Im}(z)} < 3$
- 3 Let X be binomial distributed random variable representing the number of successes after n trials. Consider the statement

$$\forall k \in [0, n], \exists \delta \in (0, 1) : P(X \leq k) > \delta.$$

Which of the following is a negation of this statement?

- (A) $\forall \delta \in (0, 1), \exists k \in [0, n] : P(X \leq k) \leq \delta$
(B) $\forall k \in [0, n], \exists \delta \in (0, 1) : P(X \leq k) \leq \delta$
(C) $\forall \delta \in (0, 1), \exists k \in [0, n] : P(X \leq k) > \delta$
(D) $\forall k \in [0, n], \exists \delta \in (0, 1) : P(X \leq k) > \delta$

- 4 Three forces of magnitudes F_1 , F_2 and F_3 act on a resting particle P on the same plane as shown in the diagram below.



NOT TO SCALE

The particle continues to remain at rest. What is the ratio $F_1 : F_2 : F_3$?

- (A) $3 : 1 : 1$ (B) $1 : 3 : 3$
 (C) $\sqrt{3} : 3 : 3$ (D) $3 : \sqrt{3} : \sqrt{3}$

- 5 Let ω be a non-real root of the equation $z^5 = 1$.

What is the value of $1 + \frac{1}{\omega} + \frac{1}{\omega^2} + \frac{1}{\omega^3} + \frac{1}{\omega^4} + \frac{1}{\omega^5}$?

- (A) -1 (B) 0
 (C) 1 (D) 2
- 6 An object of mass m is attached to a vertical spring hanging from a ceiling. The object has a displacement of y from the origin, with downwards defined as the positive direction.

When the spring is fully stretched and then released, the object is subject to a force of magnitude mky from the spring itself (where k is a positive constant) as well as gravitational force mg (where g is the acceleration due to gravity).

The net force on the object is such that it moves in simple harmonic motion.

What is the centre of this simple harmonic motion?

- (A) $\frac{g}{k}$ (B) $-\frac{g}{k}$
 (C) gk (D) $-gk$

- 7 On the complex plane, the position vector $\overrightarrow{OZ}$ is represented by the complex number z . Which of the following best describes a transformation on $\overrightarrow{OZ}$ to obtain the position vector represented by $i^3\bar{z}$?

- (A) Reflect $\overrightarrow{OZ}$ about the x -axis, then rotate anticlockwise by an angle of $\frac{\pi}{2}$ about the origin.
 (B) Reflect $\overrightarrow{OZ}$ about the y -axis, then rotate anticlockwise by an angle of $\frac{\pi}{2}$ about the origin.
 (C) Rotate $\overrightarrow{OZ}$ anticlockwise by an angle of $\frac{3\pi}{2}$ about the origin.
 (D) Reflect $\overrightarrow{OZ}$ about the x -axis, then reflect about the y -axis.

- 8 Object A is projected upwards from the ground with an initial speed of u . At the same time, object B is released from rest from height h above the ground. Both objects are subjected to gravity and air resistance proportional to the object's own velocity.

It can be shown that the general equation for the velocity of the object over time is $v = ae^{kt} + b$ for some constants a, b and k , where the direction of motion is defined as the positive direction.

Which of the following statements is correct?

- (A) $a < 0, b > 0, k > 0$ for object A and $a > 0, b < 0, k > 0$ for object B
 (B) $a < 0, b > 0, k < 0$ for object A and $a > 0, b < 0, k < 0$ for object B
 (C) $a > 0, b < 0, k > 0$ for object A and $a < 0, b > 0, k > 0$ for object B
 (D) $a > 0, b < 0, k < 0$ for object A and $a < 0, b > 0, k < 0$ for object B

- 9 Suppose that z is any complex number which lies on the unit circle in the complex plane. Which of the following inequalities is always true?

- (A) $0 \leq |z^8 - 3| \leq 2$ (B) $1 \leq |z^8 - 3| \leq 3$
 (C) $2 \leq |z^8 - 3| \leq 4$ (D) $3 \leq |z^8 - 3| \leq 5$

- 10 Which of the following integrals has the largest value?

- (A) $\int_0^\pi \cos^7 x \, dx$ (B) $\int_{-2}^2 e^{-\frac{1}{2}x^2} \, dx$
 (C) $\int_{-\sqrt{3}}^1 \tan^{-1} x \, dx$ (D) $\int_1^e \sqrt{\ln x} \, dx$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 Writing Booklet

- (a) Use the substitution $\sin \theta = \sin x - \cos x$ to evaluate 4

$$\int_0^{\frac{\pi}{2}} \frac{\cos x}{(1 + \sqrt{\sin 2x})^2} dx.$$

- (b) Let z be a complex number such that $|z + i|z|| = |z - 2i|z||$. Sketch the locus of z . 2

- (c) A trolley of mass m is initially at rest partway up a frictionless slope at an angle of θ to the horizontal. A student attempts to pull the trolley up the slope with a force of ma for some constant a . Let v and x be the trolley's speed and distance respectively at time t along the slope relative to its initial position. 2

Show that $v^2 = 2x|a - g \sin \theta|$, where g is the acceleration due to gravity.

- (d) Let $f(x)$ be a continuous function for all real x .

- (i) Show that $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$, for every $a > 0$, if and only if $f(x)$ is an even function. 4

- (ii) Suppose that $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$ for some fixed $a > 0$. 1

Show that $f(x)$ is not necessarily an even function.

- (e) It can be shown that, for any three positive numbers a, b and c , that 2

$$a + b + c \geq 3\sqrt[3]{abc}. \quad \text{(Do NOT prove this)}$$

Use this result to show that

$$\sqrt{x} + \sqrt{y} + \sqrt{z} \geq xy + yz + xz$$

where x, y and z are positive numbers such that $x + y + z = 3$.

End of Question 11

Question 12 (15 marks) Use the Question 12 Writing Booklet

- (a) Let a and b be non-zero integers.
- (i) Prove that if a^3 is even then a is even. 1
- (ii) Suppose that $(a - b)^3 + a^3 = (a + b)^3$. Prove that both a and b are even. 2
- (b) Using the result $\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$, find the solutions to 3
- $$\operatorname{cosec} \alpha = \operatorname{cosec} 2\alpha + \operatorname{cosec} 3\alpha$$
- for the domain $-\pi < \alpha \leq \pi$.
- (c) Let $\underline{n} = a\underline{i} + b\underline{j} + c\underline{k}$ be a vector that is perpendicular to a plane $\mathcal{P}$ for some constants a, b and c . Suppose that $R(x, y, z)$ is any point on $\mathcal{P}$ with position vector $\underline{r}$.
- (i) If d is the shortest distance of the plane $\mathcal{P}$ from the origin, use the relationship between the vectors $\underline{n}$ and $\underline{r}$ to show that there exists a particular position vector $\underline{r}$, which has components that satisfy 2
- $$d = \frac{|ax + by + cz|}{\sqrt{a^2 + b^2 + c^2}}.$$
- (ii) Consider the equations of two distinct planes for some constants a, b and c 1
- $$ax + by + cz = m_1$$
- $$ax + by + cz = m_2$$
- where m_1 and m_2 are distinct non-zero constants.
- Explain why the two planes are parallel to each other.
- (iii) Hence, express the shortest distance between the two parallel planes in part 2
- (ii) in terms of a, b, c, m_1 and m_2 , with consideration of different cases.
- (d) Let $I_k = \int_0^1 x^k(1 - x)^{n-k} dx$ for integers n and k , where $0 \leq k \leq n$.
- (i) Show that $I_k = \frac{k}{n - k + 1} I_{k-1}$. 2
- (ii) Hence show that $\int_0^1 \binom{n}{k} x^k(1 - x)^{n-k} dx = \frac{1}{n + 1}$. 2

End of Question 12

Question 13 (15 marks) Use the Question 13 Writing Booklet

- (a) (i) Express $\frac{8}{1+4x^4}$ as a sum of partial fractions over $\mathbb{R}$. **3**

- (ii) Hence, show that for some constant c **2**

$$\int \frac{8}{1+4x^4} dx = \ln \left(\frac{2x^2 + 2x + 1}{2x^2 - 2x + 1} \right) + 2 \tan^{-1}(2x + 1) + 2 \tan^{-1}(2x - 1) + c.$$

- (b) An object of mass m moves horizontally in the positive direction with an initial speed of u . At time t , the object has a velocity of v and is subjected to two resistance forces of magnitudes mk and $4mkv^4$, where k is a positive constant. The object decelerates until it comes to rest.

- (i) Using the result in part (a)(ii), find the time taken for the object to come to rest in terms of k and u . **2**

- (ii) Hence, find the maximum possible time that the object can take to come to rest as the initial speed is increased. Justify that this is a maximum. **2**

- (c) Suppose that $x + iy$ is a non-real root of the polynomial **2**

$$P(z) = az^4 + ibz^3 + cz^2 + idz + e$$

where a, b, c, d, e, x and y are real numbers.

By considering $P(iz)$, or otherwise, show that $-x + iy$ is also a root.

- (d) Let $\underline{r} = \left(\frac{\cos 2t}{2} \right) \underline{i} + \left(\frac{\sin 2t}{2} \right) \underline{j} + (\sin t) \underline{k}$, for some parameter t .

As t varies, $\underline{r}$ traces a curve in three-dimensional space.

- (i) All points on the curve lie on the surface of a particular sphere with a radius of 1. Find the centre of this sphere. **3**

- (ii) All points on the curve also lie on the surface of a cylinder centered about the z -axis. Find the radius of this cylinder. **1**

End of Question 13

Question 14 (15 marks) Use the Question 14 Writing Booklet

- (a) Two particles A and B move according to the following acceleration equations for some constants b and n , where $n > 0$.

$$\ddot{x}_A = -n^2(x_A - b)$$

$$\ddot{x}_B = -n^2x_B$$

Initially, particle A has a displacement of b and velocity of n , whilst particle B has a displacement of 1 at rest.

- (i) Show by integration that $\dot{x}_A^2 = n^2(1 - (x_A - b)^2)$. **2**
- (ii) Hence, or otherwise, find x_A and x_B in terms of b, n and t . **3**
- (iii) Find the range of values of b such that particles A and B never collide. **2**
- (b) The complex numbers z, z^2 and λz , where $\lambda > 1$, make an equilateral triangle on the complex plane. If $|z| = \sqrt{2}$, show that **3**

$$\lambda = \frac{1 + \sqrt{5}}{2}.$$

- (c) Suppose that $f(x)$ is a continuous function for all real x with a unique inverse function $f^{-1}(x)$.

- (i) Consider the following statement. **1**

All solutions to the equation $f(x) = f^{-1}(x)$ must satisfy $f(x) = x$.

Provide one counterexample to show that the statement is not always true.

- (ii) Now suppose that $f(x)$ is an increasing function. Prove that the statement in part (i) is true. **2**
- (iii) Hence, find all real solutions of the equation **2**

$$2\sqrt[3]{2x - 1} = x^3 + 1.$$

End of Question 14

Question 15 (15 marks) Use the Question 15 Writing Booklet

- (a) The vector equation of line ℓ_1 is $\underline{r} = \begin{pmatrix} 3 \\ 2 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$ for some parameter λ . **3**

Suppose that the line ℓ_2 passes through the point $(1, 1, 1)$ and is perpendicular to the line ℓ_1 .

Show that ℓ_2 is parallel to the position vector $\begin{pmatrix} 2 \\ 1 \\ -5 \end{pmatrix}$.

- (b) (i) Show that $\sin x < x$, for $x > 0$. **1**

- (ii) Hence, show that $\cos(\sin x) > \sin(\cos x)$ for all real x . **3**

- (c) Two identical objects A and B , each of mass m , were simultaneously launched from the ground with an initial speed of u and an angle of θ to the horizontal. Let g be the acceleration due to gravity.

Suppose that at time t , object A was subjected to air resistance of magnitudes $mk\dot{x}_A$ and $mk\dot{y}_A$ in its horizontal and vertical components respectively, where k is a positive constant, $\dot{x}_A$ and $\dot{y}_A$ are the horizontal and vertical components of object A 's speed respectively.

Object B was not subjected to any air resistance.

- (i) By integrating the relevant acceleration equations, show that the displacements equations for object A relative to the launch position are **4**

$$x_A = \frac{u \cos \theta}{k}(1 - e^{-kt}) \quad \text{and} \quad y_A = \frac{(g + ku \sin \theta)}{k^2}(1 - e^{-kt}) - \frac{gt}{k}.$$

- (ii) Hence, show that object B reaches its maximum height before object A hits the ground, but only after object A has reached its maximum height first. **4**

End of Question 15

Question 16 (15 marks) Use the Question 16 Writing Booklet

- (a) Prove that $\sqrt{3}$ is irrational. **2**
- (b) Let $P(x) = x^4 - 8x^3 + 6x^2 - 8x + 1$ and n be any non-negative integer.
- (i) Use the substitution $y = x + x^{-1}$, to express $P(x) = 0$ as a quadratic equation in terms of y . **1**
- (ii) Hence, express $P(x)$ as a product of real quadratic polynomials, in terms of x . **1**
- (iii) Find the exact value of the non-real root α with positive imaginary part, and show it has a modulus of 1. **2**
- (iv) Let $T_n = \omega^{2^n}$ be the general term of a sequence, for some root of unity ω . **1**
- For example, $T_0 = \omega, T_3 = \omega^8, T_6 = \omega^{64}$.
- Explain why there are only a finite number of possible values for T_n .
- (v) It can be shown that $\operatorname{Re}(\alpha^{2^n})$ can be written in the form $a_n - b_n\sqrt{3}$ where a_n and b_n are positive integers. **(Do NOT prove this)** **2**
- Find expressions for a_{n+1} and b_{n+1} in terms of a_n and b_n .
- (vi) Using mathematical induction, prove that $a_{n+1} > a_n$ and $b_{n+1} > b_n$ for $n \geq 0$. **3**
- (vii) By using the previous part and part (a), show that the sequence $S_n = \alpha^{2^n}$ never repeats. **2**
- (viii) Hence, explain why the following statement is FALSE. **1**
- If α is a non-real root of any monic polynomial $P(x)$ with integer coefficients, and all non-real roots of $P(x)$ have modulus 1, then α is always a root of unity.

End of paper